

Scraftie — Project README

Scraftie — Personal Music Streaming Server • Initial Build Documentation

Scraftie — Personal Music Streaming Server

A private, self-hosted personal music streaming server built from an older HP laptop.

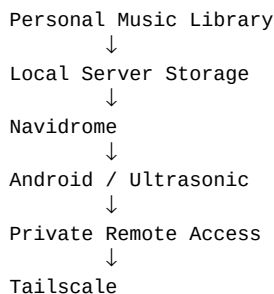
Current status: ☐ Functional initial build

Scraftie is a personal music server built incrementally from an older HP 15-da0037nr laptop.

The server runs Debian 13 (x86_64) with Navidrome providing the music-streaming service. Remote administration is performed from a Fedora workstation over SSH, while remote music access is provided through Tailscale.

The system is intended for private personal use rather than as a public streaming service.

Project Goal



The initial build focused on getting the complete pipeline working before attempting to make the server fully headless or building a custom client.

Current Status

Component	Status
Debian 13 server	☐ Completed
Music migration	☐ Completed
Navidrome	☐ Completed
SSH administration	☐ Completed
Tailscale networking	☐ Completed

Component	Status
Android streaming	☐ Verified
Cellular streaming through Tailscale	☐ Verified
Lid-close server operation	☐ Configured and tested
GNOME suspend prevention	☐ Configured
Fully headless server	☐ Planned
Custom Android client	☐ Planned
Last.fm integration	☐ Planned

Features

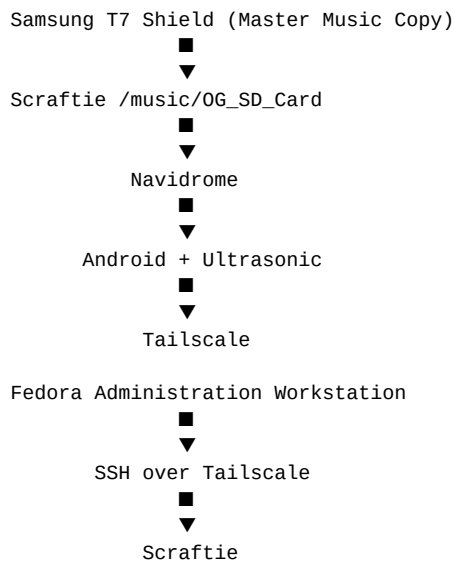
Currently implemented

- Personal music library hosted locally
- Navidrome music streaming
- Debian 13 server
- SSH administration
- Private Tailscale networking
- Android streaming through Ultrasonic
- Remote playback over cellular data
- Laptop lid configured so closing the lid does not intentionally suspend the server
- Samsung T7 Shield retained as the master music copy

Not currently implemented

- Public internet exposure
- Tailscale Funnel
- Public port forwarding
- Custom Android application
- Automated recommendation system
- Last.fm integration
- Automated upload/import workflow
- Automated offline music rotation
- Fully headless operating-system configuration

Architecture



Hardware

Model: HP 15-da0037nr

The exact CPU, RAM, storage-device model/capacity, networking hardware, and original operating system were not established in the retained build record and are intentionally not guessed here.

The Samsung T7 Shield is used as the master/original music copy.

Software Stack

Component	Role
Debian 13	Server operating system
Navidrome	Music streaming server
OpenSSH	Remote administration
Tailscale	Private remote networking
Ultrasonic	Android Navidrome/Subsonic-compatible client
Fedora	Administration workstation
Android / Motorola Edge+ 2023	Remote streaming client

Music Storage

The original music directory was renamed from:

OG SD Card

to:

```
OG_SD_Card
```

The server-side library is:

```
/music/OG_SD_Card
```

Verification:

```
find /music/OG_SD_Card -type f | wc -l
du -sh /music/OG_SD_Card
```

Results:

```
4,669 files
26G
```

The Samsung T7 Shield retains the original/master copy.

Navidrome

Configuration:

```
DataFolder = "/var/lib/navidrome"
MusicFolder = "/music/OG_SD_Card"
```

The initial service failure was:

```
Error disabling foreign_keys
unable to open database file: no such file or directory
Database could not be opened!
```

The fix was:

```
sudo mkdir -p /var/lib/navidrome
sudo chown -R navidrome:navidrome /var/lib/navidrome
sudo chown -R navidrome:navidrome /music/OG_SD_Card
sudo systemctl restart navidrome
```

Navidrome subsequently became active and scanned the library.

Remote Access

Tailscale connects Scraffie, Fedora, and the Android phone through a private network.

The real Tailscale address is intentionally omitted:

```
ssh cprquecian@<SCARFTIE_TAILSCALE_IP>
```

Tailscale Funnel is not part of the intended architecture.

Android Verification

The Motorola Edge+ 2023 used Ultrasonic to connect to Navidrome.

The decisive remote test was performed with Wi-Fi disabled. Music continued playing over cellular data through Tailscale.

This demonstrated actual remote connectivity rather than merely local-Wi-Fi playback.

Lid and Power Management

The server initially stopped responding over Tailscale when the laptop lid was closed.

The intended `/etc/systemd/logind.conf` settings became:

```
HandleLidSwitch=ignore
HandleLidSwitchExternalPower=ignore
HandleLidSwitchDocked=ignore
LidSwitchIgnoreInhibited=yes
```

GNOME did not contain the expected `lid-close-ac-action` key.

Available keys were inspected with:

```
gsettings list-keys org.gnome.settings-daemon.plugins.power
```

The AC inactivity action was changed with:

```
gsettings set org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type 'nothing'
```

Verification:

```
gsettings get org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type
```

Result:

```
'nothing'
```

Security Model

- Personal/private use

- SSH administration
- Tailscale private networking
- No intentional public port forwarding
- No Tailscale Funnel
- Credentials excluded from GitHub
- No secrets committed to the repository

Documentation

- docs/installation.md — chronological initial build
- docs/architecture.md — current architecture and service model
- docs/troubleshooting.md — failures, diagnosis, fixes, and verification
- docs/roadmap.md — future work

Lessons Learned

1. A successfully installed application can still fail as a systemd service.
2. Service accounts need access to their data and application directories.
3. Laptop hardware introduces server-specific power-management problems.
4. Remote access should be tested from outside the local network.
5. Configuration should be inspected on the actual installed system instead of assumed.
6. Troubleshooting history is valuable engineering documentation.

Current Limitations

- GNOME remains installed.
- The server is not yet fully headless.
- The custom Android client does not yet exist.
- Recommendation and listening-analysis features are not implemented.
- Last.fm integration is not implemented.
- Automated music import workflows are not implemented.
- Public-facing streaming is not intended.